

AI LEAD SCORING AND ESTIMATED LEAD VALUE PREDICTION

Machine Learning – Supervised Regression Project

1. Project Overview

The AI Lead Scoring and Estimated Lead Value Prediction project is a Supervised Machine Learning Regression project. The main objective is to predict the Estimated Lead Value of a potential customer based on the characteristics and interactions associated with the lead.

Historical lead data is used to identify relationships between lead-related features and expected monetary value. Multiple regression algorithms were evaluated, and Linear Regression was selected as the final model based on the highest initial R^2 score.

2. Problem Statement

Businesses generate leads through websites, Google, Facebook, LinkedIn, referrals, and other marketing channels. Each lead can have a different potential value. This project develops a machine learning model to estimate the value of a new lead.

- Analyze lead-related information.
- Identify important features influencing lead value.
- Predict Estimated Lead Value for a new lead.
- Help businesses prioritize valuable leads.
- Support sales and marketing teams with data-driven decisions.

3. Machine Learning Type

Machine Learning Type: Supervised Learning

Problem Type: Regression

Target Variable: Estimated_Lead_Value

Regression is appropriate because the target variable is a continuous numerical value.

4. Dataset

The dataset contains historical lead information. Numerical and categorical features were initially considered for predicting Estimated_Lead_Value.

- Website visits
- Time spent on website
- Email interactions
- Previous interactions
- Lead quality

- Annual income
- Sales calls
- Recent lead activity
- Lead source
- Industry

Target variable: Estimated_Lead_Value

5. Data Preprocessing

1. Loaded the dataset using Pandas.
2. Identified independent and dependent variables.
3. Handled missing values where required.
4. Converted categorical variables into numerical/dummy variables.
5. Separated Estimated_Lead_Value from the independent variables.
6. Performed feature selection to reduce input variables.
7. Applied StandardScaler to standardize the selected features.

6. Feature Selection

SelectKBest with the f_regression scoring function was used to reduce the number of input variables. The final model used the following eight selected features:

- Total_Visits
- Time_Spent_Website
- Email_Clicks
- Previous_Interactions
- Lead_Quality_Score
- Annual_Income
- Sales_Calls
- Days_Since_Last_Activity

The same eight features must be supplied in the deployment phase so that the input matches the feature structure used during model training.

7. Model Development

The following supervised regression algorithms were evaluated:

Algorithm	Initial R ² Score
Linear Regression	0.9475
SVM – Linear	0.0893
SVM – Non-Linear	0.0010
Decision Tree Regression	0.6618

Linear Regression achieved the highest score in the initial model comparison and was selected as the final regression algorithm.

8. Final Model Creation

The final model was trained using the selected eight features. StandardScaler was fitted on these features before training Linear Regression. The trained Linear Regression model and the fitted scaler were saved using Pickle.

Saved files:

- Lead_Scoring_Linear_Regression_8Features.pkl
- Lead_Scoring_Scaler_8Features.pkl

9. Model Evaluation

Model performance was evaluated using the R^2 score. The initial Linear Regression comparison produced an R^2 score of approximately 0.9475, which was higher than the other tested algorithms.

For final academic evaluation, a test-set R^2 score is recommended because it measures performance on unseen data rather than only on the training data.

10. Deployment Phase

During deployment, the saved model and scaler are loaded. A new lead is entered using the same eight selected features, the new input is transformed using the saved scaler, and the transformed data is passed to the saved Linear Regression model.

Example deployment input:

- Total_Visits = 15
- Time_Spent_Website = 650
- Email_Clicks = 5
- Previous_Interactions = 12
- Lead_Quality_Score = 75
- Annual_Income = 45000
- Sales_Calls = 10
- Days_Since_Last_Activity = 8

Example prediction:

Estimated Lead Value: 113718.26

For the example input, the model estimated a lead value of approximately 113,718.26.

11. Multiple Lead Prediction

The same trained model can predict multiple leads at once. Each input row represents one lead, and each value in the prediction array corresponds to that lead.

- prediction[0] → first lead

- prediction[1] → second lead
- prediction[2] → third lead
- prediction[5] → sixth lead
- prediction[8] → ninth lead

Therefore, prediction[5] or prediction[8] does not represent a different model. It refers to the corresponding row in the batch of leads supplied for prediction.

12. Complete Project Workflow

1. Historical Lead Dataset
2. Data Cleaning
3. Data Preprocessing
4. Encoding Categorical Variables
5. Feature Selection
6. SelectKBest – 8 Features
7. Train/Test Split
8. StandardScaler
9. Regression Algorithm Comparison
10. Linear Regression Selection
11. Final Model Training
12. Model and Scaler Saving
13. Deployment
14. New Lead Input
15. Feature Scaling
16. Estimated Lead Value Prediction

13. Business Benefits

- Identifies leads with higher potential monetary value.
- Helps prioritize sales follow-ups.
- Improves sales-team efficiency.
- Supports effective marketing resource allocation.
- Reduces manual lead evaluation.
- Supports data-driven decision-making.

- Automatically estimates the potential value of new leads.

14. Conclusion

The AI Lead Scoring and Estimated Lead Value Prediction project demonstrates the application of Supervised Machine Learning Regression to a real-world business problem.

Linear Regression, Linear SVM, Non-Linear SVM, and Decision Tree Regression were compared. Linear Regression achieved the best initial performance, with an R^2 score of approximately 0.9475.

Feature selection reduced the final model inputs to eight features. The final Linear Regression model and scaler were saved using Pickle and used during deployment to predict Estimated Lead Value for new leads.

15. Final Project Statement

The objective of this project is to develop a supervised machine learning regression model that predicts the Estimated Lead Value of potential customers using important lead characteristics. After comparing multiple regression algorithms and performing feature selection, Linear Regression was selected as the final model and deployed using eight selected features to generate estimated lead values for new leads.